



Course: Database Administration

Lecturer: Mr. Thear Sophal

Week 2 – Practical Session

The review of SQL

Basic Queries

1. Retrieve all data from the Airlines table.
2. Retrieve the names of all airlines based in the USA.
3. Retrieve the names of all airports located in New York.
4. Retrieve the number of flights for each airline.
5. Retrieve the details of the flight with FlightID 101.

Aggregation Functions

6. Find the total number of flights in the database.
7. Calculate the average departure time of flights.
8. Find the airline with the most flights.
9. Find the airport with the highest number of outgoing flights.
10. Calculate the total number of passengers.

Joins

11. Retrieve the details of flights along with the name of the airline.
12. Retrieve the details of tickets along with the passenger's name.
13. Retrieve the details of flights, including the names of the source and destination airports.
14. Retrieve the names of passengers who booked a ticket for a flight departing from New York.
15. Retrieve the names of passengers who booked a ticket with Emirates.

Subqueries

16. Retrieve the details of passengers who booked a ticket with Emirates.
17. Retrieve the details of flights departing from New York.
18. Retrieve the names of passengers who booked a ticket for a flight departing from New York with Emirates.
19. Retrieve the details of flights where the departure time is later than the average departure time.
20. Retrieve the names of passengers who booked a ticket for a flight operated by the airline with the fewest flights.

Create View

21. Create a view named "**FlightDetails**" that retrieves the details of flights along with the names of the source and destination airports.
22. Query the **FlightDetails** view to retrieve the flight details.

Stored Procedure

23. Create a stored procedure named "**GetFlightsByAirline**" that takes an airline name as input and retrieves all flights operated by that airline.
24. Execute the stored procedure to retrieve flights operated by Emirates.
`CALL GetFlightsByAirline('Emirates');`
25. Create a stored procedure named "**GetPassengerDetails**" that takes a passenger name as input and retrieves all details of that passenger.
26. Execute the stored procedure to retrieve details of the passenger named "John Smith".